Exact Deletion Limits and Realizability Certificates for the Crossing Number of Complete Graphs

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Abstract

The supplied manuscript claims the Harary–Hill formula for ordinary crossing numbers by combining a corrected cycle-intersection pairing, an even-node order detector, polynomial injectivity, and a symmetric isotropic dimension argument. We map those mechanisms without replacing actual drawings by unconstrained rotation matrices. The local study proves the exact limit of all nonnegative single-vertex deletion certificates, including an integer four-uniform relaxation attaining that limit. It retains the odd-order correction, derives quantitative consequences for near-optimal drawings under explicitly stated minor bounds, and constructs a seven-vertex count model which passes every single-deletion bound but fails a smaller planar obstruction. General induced-subgraph and signed support ledgers clarify what extra information a stronger certificate must retain. Original Ram frame results are used only to control finite arithmetic and declared coordinate norms, not to supply topological realizability. Exact finite tests accompany the proofs; neither tests nor the source's publication are treated as an independent global verification. Contribution assessment: C2/5, the report provides useful but limited exact deletion-optimality and realizability diagnostics, without a new universal crossing bound or a substantial advance beyond classical counting and finite optimization.

1 The problem and the evidential boundary

An ordinary drawing places distinct vertices at distinct plane points and represents each edge by a simple continuous arc whose interior avoids all vertices. Intersections of edge interiors are finitely many proper double crossings; tangencies, overlaps, and triple crossings are excluded. The crossing count $c(D)$ counts *points*, including repeated crossings of the same edge pair. The ordinary crossing number is the minimum over all such drawings. It is not the rectilinear crossing number, a two-page minimum, an odd-crossing count, or a count of intersecting edge pairs.

Write

$$Z(n) = \frac{1}{4}\left\lfloor \frac{n}{2} \right\rfloor \left\lfloor \frac{n-1}{2} \right\rfloor \left\lfloor \frac{n-2}{2} \right\rfloor \left\lfloor \frac{n-3}{2} \right\rfloor. \tag{1}$$

For $s \geq 2$,

$$Z(2s) = \frac{s(s-1)^2(s-2)}{4}, \qquad Z(2s+1) = \frac{s^2(s-1)^2}{4}. \tag{2}$$

The supplied September 23 manuscript [1] claims $\mathrm{cr}(K_n) = Z(n)$ for every n. We distinguish its global proof from the local results below. No conclusion about unrestricted drawings follows merely from a successful finite detector calculation or an explicit construction attaining $Z(n)$.

A *normalized* drawing here means polygonal, straight near vertices and crossings, and without crossings between adjacent edges. Repeated crossings between independent edges remain permitted. This is deliberately the source's convention, weaker than the frequently used *good drawing* convention which also permits at most one crossing per pair. All deletion statements below concern a specified normalized drawing, or an explicitly defined four-uniform count model. An induced drawing $D[S]$ retains the original routes on the vertex set S; it is not reoptimized after deletion.

The comparison TR-item-071 [7] concerns $K_{m,n}$. Its rotation frequencies, diagonal self-star correction, and one-part survival factor $n-2$ cannot be transplanted to K_n, where an independent crossing uses four vertices and survives $n-4$ one-vertex deletions. Original proof passages in TR248 [8] were also freshly read. Their finite frame bounds concern a named vector space and physical norm; they do not produce a drawing from a feasible matrix.

2 The supplied proof, with its mechanisms retained

2.1 Normalization is a geometric premise

The source's appendix starts with a drawing having minimum integer crossing count. It chooses finitely many disjoint disks at vertices and crossings, but it does *not* assume a continuous arc meets a disk boundary only finitely often. Instead it removes the passage between first and last visits inside a parameter neighborhood. The retained pieces are compact and disjoint, so disjoint exterior neighborhoods admit polygonal replacements. Straight chords at crossing disks and spokes at vertices produce no more crossings. Exchanging the prefixes of two adjacent crossing edges, then erasing loops in the resulting walks, would reduce the count. This gives the claimed no-adjacent-crossing normalization.

The compactness, endpoint, and local routing assertions are actual plane geometry obligations. An arbitrary cyclic order at each vertex, even with all relevant matrices nonsingular, does not establish them. The entire normalization argument was read, not executed as a finite algorithm for every continuous drawing.

2.2 The corrected cycle pairing

Fix edge orientations. The edge space is real, $[pi] = -[ip]$, $[ii] = 0$, and $\partial[ip] = e_p - e_i$. Cycles are the kernel of this boundary map. Let $I(e, f)$ be the signed sum of interior crossings. At each vertex w, choose a gap in the circular order of outward spokes and let $T_w(e, f)$ equal 1 when e precedes f, -1 when the order is reversed, and zero when they agree. Let σ_{we} be 1 for an edge directed out of w and -1 for an edge directed into w. The source form is

$$J(e, f) = I(e, f) + \frac{1}{2} \sum_{w \in e \cap f} \sigma_{we}\sigma_{wf}T_w(e, f). \tag{3}$$

This is skew-symmetric, and the source proves

$$J(z, z') = 0 \qquad (\partial z = \partial z' = 0). \tag{4}$$

The shared-vertex term is not optional. The proof separates tracks of two cycles by left and right perturbations, averages their intersection counts, and uses cancellation of signed intersections of closed plane walks. In a vertex disk the oriented chord identity is

$$I(C, E) = \frac{1}{2} \sum_{u \in \partial C,\, v \in \partial E} \epsilon_C(u)\epsilon_E(v)T(u, v).$$

Coincident spokes contribute zero only after the two perturbations are averaged. Reversing a traversal changes its coefficient, not this averaging. This explains both the half factor and the endpoint signs in (3).

2.3 The even-node detector

For distinct real nodes $u_1, \dots, u_N$ and $G(t) = \prod_p (t - u_p)$, coefficient comparison in Lagrange interpolation gives

$$\sum_p \frac{A(u_p)}{G'(u_p)} = 0 \quad (\deg A \le N - 2). \tag{5}$$

For $N = 2s$, the order of the nodes below is a prescribed spoke order and need not be numerical order. Put

$$L_s = \{(p, q) : 1 \le q < p \le s\} \cup \{(p, q) : s < p < q \le 2s\}.$$

The source proves that evaluation on this set is invertible for each rectangle $\mathcal{P}_{s-1,s-2}$ and $\mathcal{P}_{s-2,s-1}$ of separately bounded bivariate degrees. In the first rectangle the rows $s, s + 1$ force factors $(u - u_s)(u - u_{s+1})$; deleting those nodes leaves the *other* rectangle on L_{s-1}. The second case uses columns $1, 2s$ instead. Both have dimension $s(s - 1)$, and the two cases induct simultaneously.

If $S(p, q)$ is one half times the order sign, the bilinear form

$$M(A, B) = \sum_{p,q} S(p, q) \frac{u_p - u_q}{G'(u_p)G'(u_q)} A(u_p, u_q)B(u_p, u_q) \tag{6}$$

is nonsingular between the two rectangles. Indeed, (5) allows adding a row term and a column term to S. The resulting off-diagonal array is -1 on the first triangle of L_s, 1 on the second, and zero elsewhere. In the two evaluation coordinates the form is diagonal with nonzero entries. These signs are not a claim of positive definiteness. This argument is symbolic for every prescribed order and all distinct nodes; checking finitely many matrices is supplemental evidence only.

2.4 Injectivity and isotropy are separate steps

For $n = 2s + 1$, choose distinct labels t_i and weights $w_i = \prod_{j \ne i} (t_i - t_j)^{-1}$. Polynomially weighted edge chains are cycles, so (4) implies

$$\sum_{i,p,k,q} J([ip], [kq]) w_i w_p w_k w_q H(t_i, t_p, t_k, t_q) = 0 \tag{7}$$

whenever every separate degree of H is at most $n - 2$. Let $\mathcal{S}$ consist of polynomials $P(x, u, y, v)$ separately of degree at most $s - 2$, symmetric within (x, u) and within (y, v), but not necessarily under exchange of the two whole pairs. Its dimension is $\binom{s}{2}^2$.

The evaluation coordinates are *ordered* pairs of undirected, independent edges with nonzero signed crossing count; call this set $\mathcal{T}$. The source's injectivity argument applies (7) to

$$H = P(x, u, y, v)(u - v)(u - y)(v - x)U(x)V(y)B(u, v),$$

where $\deg U, \deg V \le s$ and $B \in \mathcal{P}_{s-1,s-1}$. The degree ceiling is exactly $2s - 1 = n - 2$. The kernel assumption removes independent-edge terms; the three differences remove all shared endpoints except $i = k$. Products UV span degree $2s$ and can localize at one of the $2s + 1$ nodes. The detector

forces $P(t_i, u, t_i, v) = 0$ identically. A univariate root count and the two endpoint symmetries then show that

$$C = (x - y)(x - v)(u - y)(u - v)$$

divides every kernel polynomial. Dividing by C preserves the kernel, the symmetries, and admissible separate degrees, while lowering each degree by two. A minimum-degree nonzero kernel element is therefore impossible.

Next the full six-difference product $\Delta = (x - u)(y - v)C$ gives, for every $P, Q \in \mathcal{S}$,

$$E(P)^T A E(Q) = 0, \qquad A = \operatorname{diag}(a_\alpha)_{\alpha \in \mathcal{T}}, \qquad a_\alpha \neq 0. \tag{8}$$

Each a_α includes J, four node weights, and Δ. Reversing one edge changes both J and Δ, so the four endpoint-orientation choices agree. Exchanging the two whole edges gives the opposite diagonal coefficient. Thus A is a *symmetric nondegenerate* form with paired positive and negative entries; it is not the skew form J. If $W = E(\mathcal{S})$, then $W \subset W^\perp$ gives $2 \dim W \leq |\mathcal{T}|$. Injectivity is needed to replace $\dim W$ by $\dim \mathcal{S}$. Finally $c(D) \geq |\mathcal{T}|/2$ counts unordered supported pairs, not half of the actual crossing points with multiplicity. This is the source's global odd-order lower-bound chain, retained with attribution, not credited here as a new Ram theorem.

2.5 The matching upper construction

The source uses the established endpoint-sum two-page construction: vertices $0, \ldots, n - 1$ lie on a line; the page of ij is determined by whether $i + j \bmod n$ belongs to the first $\lfloor n/2 \rfloor$ residues. Same-page semicircles cross precisely when endpoints alternate. A generic choice of increasing spine positions avoids triple crossings without changing this alternation test. In the cyclic gap count, multiplication by two permutes residues only at odd n. At even n the source separately balances parity classes of endpoint sums. The final count is

$$c(D) = \frac{1}{4} \sum_{k=2}^{n-2} (k - 1)(n - k - 1)\big(n - 2 \min(k, n - k)\big) = Z(n). \tag{9}$$

The telescoper is $B_k = k(k-1)(n-k-1)(n-k-2)$, with $B_k - B_{k-1} = 2(k-1)(n-k-1)(n-2k)$. This gives a genuine drawing and an upper bound. It does not by itself prove the unrestricted lower bound. No correction to these scoped source mechanisms was established in this study.

3 Exact deletion and the parity correction

Each crossing of a normalized drawing uses exactly four distinct vertices. Treat different crossing points as distinct even when their endpoint four-sets agree. For $S \subset [n]$, a crossing survives in $D[S]$ precisely when its four endpoints are in S. Consequently, for $4 \leq m \leq n$,

$$\sum_{\substack{S \subset [n] \\ |S| = m}} c(D[S]) = \binom{n-4}{m-4} c(D). \tag{10}$$

This is an exact double count and requires neither minimality nor the Harary–Hill assertion. In particular, for $n > 4$,

$$\sum_{v} c(D - v) = (n - 4) c(D). \tag{11}$$

If adjacent-edge crossings had not been excluded, writing $a(D)$ for their number would instead give $(n-4)c(D)+a(D)$: those crossings use three vertices and survive $n-3$ deletions. This is why the geometric normalization cannot be omitted from the transfer.

Put $\delta_n(D) = c(D) - Z(n)$, allowing this integer to be negative unless a lower bound has independently been assumed or proved. Substitution yields the exact identity

$$(n-4)\delta_n(D) = \sum_v \delta_{n-1}(D-v) - \tau_n, \qquad \tau_n = (n-4)Z(n) - nZ(n-1). \tag{12}$$

Direct use of (2) gives

$$\tau_{2s} = 0, \qquad \tau_{2s+1} = \frac{s(s-1)^2}{2} = \frac{(n-1)(n-3)^2}{16}. \tag{13}$$

For example, the odd computation is

$$\frac{s(s-1)^2}{4}\big[(2s-3)s - (2s+1)(s-2)\big] = \frac{s(s-1)^2}{2}.$$

Thus nonnegative deletion defects suffice at even order but not at odd order. This is a complete-graph calculation, not the bipartite correction from TR071 with the letters renamed.

Corollary 3.1 (Exact one-step information)**.** *If every $D-v$ has at least $Z(n-1)$ crossings, then*

$$c(D) \geq \left\lceil \frac{nZ(n-1)}{n-4} \right\rceil = \begin{cases} Z(n), & n \text{ even}, \\ Z(n) - \lfloor \tau_n/(n-4) \rfloor, & n \text{ odd}. \end{cases}$$

At odd order, proving $\sum_v \delta_{n-1}(D-v) \geq \tau_n$ would restore the exact target. That assertion is additional information, not a consequence of nonnegativity.

Similarly, if all smaller orders satisfy the target, a least-order counterexample must have odd order. This familiar parity reduction is recorded for accuracy, not asserted as a novel theorem.

For general m, (10) becomes

$$\binom{n-4}{m-4}\delta_n(D) = \sum_{|S|=m} \delta_m(D[S]) - \Gamma_{n,m}, \tag{14}$$

$$\Gamma_{n,m} = \binom{n-4}{m-4}Z(n) - \binom{n}{m}Z(m).$$

The normalized target ratios, for $n \geq 4$, are

$$\frac{Z(n)}{\binom{n}{4}} = \begin{cases} \dfrac{3(n-2)(n-4)}{8(n-1)(n-3)}, & n \text{ even}, \\ \dfrac{3(n-1)(n-3)}{8n(n-2)}, & n \text{ odd}. \end{cases}$$

Successive odd-to-even ratios agree. Direct subtraction at the preceding even-to-odd step gives

$$\frac{Z(2s+1)}{\binom{2s+1}{4}} - \frac{Z(2s)}{\binom{2s}{4}} = \frac{3(s-1)}{(2s-3)(2s-1)(2s+1)} > 0 \qquad (s \geq 2). \tag{15}$$

The ratios are therefore nondecreasing and $\Gamma_{n,m} \geq 0$. This nonzero budget explains why induced-subgraph bounds of a fixed order need not reach the full target.

4 The exact limit of weighted single deletions

Fix $n \geq 5$ and nonnegative real numbers L_v that have been certified as lower bounds for $c(D-v)$ for the particular class being considered. Consider nonnegative weights λ_v satisfying

$$\sum_{v \notin T} \lambda_v \leq 1 \qquad \text{for every four-set } T \subset [n]. \tag{16}$$

Each actual crossing then contributes total weight at most one to the weighted sum of its surviving subdrawings. Hence $c(D) \geq \sum_v \lambda_v L_v$. Condition (16) is required on all four-sets because no extra support or realizability information has been supplied.

Theorem 4.1 (Optimal nonnegative single-deletion certificate)**.** *For $k = n-4$ and $L_v \geq 0$,*

$$\max_{\lambda \geq 0,\ (16)} \sum_v \lambda_v L_v = \max\left\{ \max_v L_v,\ \frac{1}{k} \sum_v L_v \right\}. \tag{17}$$

Thus arbitrary tuning of these weights cannot improve both the strongest single-subdrawing bound and uniform averaging.

Proof. Relabel jointly so $\lambda_1 \geq \cdots \geq \lambda_n \geq 0$, and set $\lambda_{n+1} = 0$, $a_j = \lambda_j - \lambda_{j+1} \geq 0$. The coverage constraint is equivalent to the sum of the largest k weights being at most one, that is,

$$\sum_{j=1}^{n} \min(j,k) a_j \leq 1.$$

On the other hand $\sum_v \lambda_v L_v = \sum_j a_j \sum_{v \leq j} L_v$. For $j \leq k$ the inner sum is at most $j \max_v L_v$; for $j \geq k$ it is at most $\sum_v L_v$, using nonnegativity. The objective is therefore at most the right side of (17) times the displayed coverage budget. Equality is attained either by weight one at a maximizing vertex and zero elsewhere, or by $\lambda_v = 1/k$ for every vertex. □

Negative proposed lower bounds should first be replaced by zero, which is always a valid crossing lower bound. Signed combinations that exploit additional equalities are outside this nonnegative-coverage theorem.

Theorem 4.2 (Integer sharpness without drawing realizability)**.** *Let all L_v be nonnegative integers. In the class of multisets of four-sets of $[n]$, count each four-set occurrence as one formal crossing. The least total number C for which deleting each vertex v leaves at least L_v occurrences is exactly*

$$C_* = \max\left\{ \max_v L_v,\ \left\lceil \frac{\sum_v L_v}{n-4} \right\rceil \right\}. \tag{18}$$

Consequently, integrality alone cannot improve this bound in that four-uniform model. Such a multiset need not be a plane drawing.

Proof. The lower bound follows from one deletion and from the sum of deletions. For $C = C_*$ set capacities $b_v = C - L_v$. They lie in $[0, C]$ and have total $nC - \sum_v L_v \geq 4C$. Choose integers $0 \leq d_v \leq b_v$ summing to $4C$, by filling capacities until that sum is reached.

We now realize these d_v as degrees of a four-uniform multihypergraph with exactly C edge occurrences. If $C = 0$, everything is empty. For $C > 0$, select four vertices of largest positive degree and decrement each by one, recording them as a four-set. There are at least four positive degrees, since their sum is $4C$ and each is at most C. Every degree equal to C is selected: at most four can have that value, and they are largest. The remaining degrees are nonnegative, at most $C-1$, and sum to $4(C-1)$. Induction completes the construction. Vertex v belongs to d_v occurrences, so its deletion leaves $C - d_v \geq L_v$, as required. □

This constructive sharpness statement is stronger than merely solving the linear weight problem, but it is sharpness in a deliberately relaxed combinatorial model. It does not establish sharpness in the set of actual drawings. Adding finite-rank, moment, or Gram coordinates without new realizability constraints does not change that distinction.

5 Worked obstruction and finite-order corrections

5.1 Seven formal crossings that cannot be a drawing

On vertices $\mathbb{Z}/7\mathbb{Z}$ take the seven four-sets

$$T_i = \{i, i+1, i+2, i+3\} \pmod 7, \qquad i = 0, \dots, 6. \tag{19}$$

Each vertex belongs to four sets. Deleting any vertex leaves exactly three occurrences, so this model satisfies all seven bounds $L_v = Z(6) = 3$ with $C = 7$. Formula (18) indeed gives $C_* = 7$, whereas $Z(7) = 9$. This is not a counterexample drawing.

In fact $S = \{0, 1, 3, 4, 6\}$ contains none of the seven four-sets. If these occurrences were the full crossing data of an actual normalized K_7 drawing, its induced K_5 on S would have no crossings. But a simple planar graph with five vertices has at most $3 \cdot 5 - 6 = 9$ edges, whereas K_5 has ten. The planar bound follows from Euler's formula and the fact that each face in a simple connected plane graph with at least three vertices has boundary length at least three, with bridges counted on both sides. Thus the formal model fails a smaller topological constraint. No use of the full Harary–Hill formula is needed for this contradiction.

The example shows why passing every single-deletion lower bound, even integrally and with exact incidence sums, does not certify a common drawing. It does not claim that one additional K_5 inequality alone closes every seven-vertex relaxation.

5.2 The exact odd loss

For constant input $L_v = Z(n-1)$, the exact relaxed bound and its loss are:

n	$Z(n-1)$	$Z(n)$	C_*	$Z(n) - C_*$
5	0	1	0	1
7	3	9	7	2
9	18	36	33	3
11	60	100	95	5
13	150	225	217	8
15	315	441	430	11

Every row follows from exact integers and (13). Theorem 4.2 proves that no combination of these one-deletion lower bounds, the four-endpoint incidence model, and integrality can remove the displayed loss. Extra geometric information, higher induced-subgraph constraints, or the source's universal signed-pairing argument is needed.

For heterogeneous inputs at $n = 7$, bounds $(4, 4, 4, 4, 3, 3, 3)$ give total 25 and therefore $c(D) \geq \lceil 25/3 \rceil = 9$. Bounds $(4, 4, 4, 3, 3, 3, 3)$ give only eight. If a single L_v exceeds the averaging bound, its own induced drawing controls the answer, exactly as (17) predicts. These examples assume the listed individual bounds have been proved; they do not manufacture drawings attaining them.

6 Near-optimality statements with explicit premises

Let a normalized drawing have $c(D) = Z(n) + h$ with $h \in \mathbb{Z}_{\geq 0}$. Assume also, independently, that every $D - v$ has at least $Z(n-1)$ crossings, and set $e_v = \delta_{n-1}(D - v) \in \mathbb{Z}_{\geq 0}$. Then

$$\sum_v e_v = (n-4)h + \tau_n. \tag{20}$$

Hence for any integer $q \geq 1$,

$$\#\{v : e_v \geq q\} \leq \min\left(n, \left\lfloor \frac{(n-4)h + \tau_n}{q} \right\rfloor\right). \tag{21}$$

This is just counting nonnegative integer slack, but the exact parity budget matters. At even order, $h = 0$ forces every single deletion to attain the target. With $h = 1$, at least four deletions attain the target. At odd order, even $h = 0$ leaves the positive budget τ_n, so the same hereditary-optimality assertion is false without new input.

For each vertex let d_v count crossing points involving that vertex, including multiplicities. Then

$$d_v = c(D) - c(D - v) = Z(n) - Z(n-1) + h - e_v, \qquad \sum_v d_v = 4c(D). \tag{22}$$

At even n, put $\kappa = 4Z(n)/n = Z(n) - Z(n-1)$. We obtain

$$d_v \leq \kappa + h, \qquad \sum_v |d_v - \kappa| \leq nh + \sum_v e_v = (2n-4)h. \tag{23}$$

Thus exact even-order optimality under the assumed deletion bounds forces uniform crossing incidence at every vertex. This is a combinatorial incidence statement, not geometric closeness to a two-page drawing.

These statements can be checked on a concrete drawing without assuming the global conjecture, if its induced counts are known exactly. To apply them to every drawing at an arbitrary order, the lower-bound premise must be supplied by a genuine theorem, not by the name of the target Z.

7 Higher-order information and signed support losses

7.1 General nonnegative induced-subgraph certificates

Let $\mathcal{F}$ be a specified family of vertex subsets of sizes at least four. For each $S \in \mathcal{F}$, suppose $c(D[S]) \geq L_S$ has been established. If $\lambda_S \geq 0$ and

$$\sum_{\substack{S \in \mathcal{F} \\ T \subset S}} \lambda_S \leq 1 \quad \text{for every four-set } T,$$

then

$$c(D) \geq \left\lceil \sum_{S \in \mathcal{F}} \lambda_S L_S \right\rceil. \tag{24}$$

The proof counts each crossing occurrence with its total coverage weight. Uniform weights $1/\binom{n-4}{m-4}$ recover (10). Nonuniform higher-order certificates can exploit additional data, but the family must be certified on actual induced drawings or as a valid relaxation containing all such drawings. Checking only sampled subsets cannot justify constraints for the missing ones.

Second-order deletion identities retain information not visible in the single-deletion sum. For distinct u, v define

$$d_{uv} = c(D) - c(D-u) - c(D-v) + c(D-\{u,v\}).$$

Inclusion–exclusion shows that d_{uv} is exactly the number of crossing points involving both vertices, so $d_{uv} \geq 0$ and $\sum_{u<v} d_{uv} = 6c(D)$. Also

$$\sum_v d_v^2 = \sum_{\xi,\eta} |V(\xi) \cap V(\eta)|,$$

where ξ, η range over ordered crossing occurrences, including $\xi = \eta$, and $V(\xi)$ is their four-vertex support. Expanding both sides as sums of indicators proves the formula. These are useful overlap constraints; neither their positivity nor a finite moment matrix is a sufficient drawing-realization criterion.

7.2 A conditional signed-support defect ledger

For each unordered independent edge pair b, let m_b be its number of crossing points and j_b its signed intersection count in fixed orientations. Define

$$N_{\text{sup}} = \#\{b : j_b \neq 0\}, \qquad W_{\text{loss}} = \sum_b \big(m_b - 1_{\{j_b \neq 0\}}\big) \geq 0.$$

Then the exact identity

$$c(D) = N_{\text{sup}} + W_{\text{loss}} \tag{25}$$

records what is lost in the source's passage from crossings to nonzero signed support. If the source's injectivity and isotropy premises have been established for this drawing at odd order, they give $N_{\text{sup}} \geq Z(n)$, and hence $W_{\text{loss}} \leq c(D) - Z(n)$.

Let $U = \#\{b : m_b > 0\}$. The repeat count beyond the first crossing on a pair is $R = \sum_{m_b>0}(m_b - 1) = c(D) - U$, and let $Z_0 = \#\{b : m_b > 0,\ j_b = 0\}$. Then $W_{\text{loss}} = R + Z_0$. Each zero-signed active pair has at least two crossings, so it contributes at least two to W_{loss}. Consequently near-optimality $c(D) \leq Z(n) + h$ under the stated source premises gives $R \leq h$ and $Z_0 \leq \lfloor h/2 \rfloor$. At $h = 0$, each active pair has exactly one crossing and nonzero sign. This is a quantitative extraction from the source support bound, not a new proof of that bound or a classification of optimal drawings.

8 Finite Ram certificates and their exact domains

The freshly read original TR248 baseline establishes two facts relevant here. A finite family of measurements cannot have a positive lower frame bound on an infinite-dimensional Hilbert space: the orthogonal complement of its finitely many Riesz representatives contains a nonzero invisible vector. In a specified finite model, however, an operator error $\|\widetilde{E} - E\|_2 \leq \eta$ gives

$$\sigma_{\min}(E) \geq \sigma_{\min}(\widetilde{E}) - \eta. \tag{26}$$

The proof is the triangle inequality on every unit input vector. Entrywise enclosures can provide η through the Frobenius norm. The original text also requires a physical Gram matrix when coefficient coordinates are not orthonormal. These are classical results transferred with attribution, not new crossing-number arguments.

For the source's finite polynomial evaluation map, one may declare the Euclidean norm of coefficients in a named basis, or a positive polynomial Gram norm and the corresponding mass

scaling. The choice affects conditioning and residual tolerances, but not exact rank. In contrast, the signed form A in (8) is indefinite and is *not* a positive sampling measure. Replacing it by $|A|$, or applying an ordinary positive-semidefinite argument to A, loses the isotropy identity.

Proposition 8.1 (A finite residual threshold for the source's inertia step)**.** *Let A be a real diagonal $2p$ by $2p$ matrix with exactly p positive and p negative entries, each of absolute value at least $a > 0$. Let $E : \mathbb{R}^d \to \mathbb{R}^{2p}$ be an actual full-column-rank matrix in specified Euclidean coordinates. If*

$$\|E^T A E\|_2 < a\,\sigma_{\min}(E)^2, \tag{27}$$

then $d \le p$.

Proof. If $d > p$, the map from $\mathbb{R}^d$ to the p negative coordinates of Ex has a nonzero kernel vector. Normalize it to $\|x\|_2 = 1$. Then Ex has only positive coordinates, whence $x^T E^T A E x \ge a\|Ex\|_2^2 \ge a\sigma_{\min}(E)^2$. This contradicts (27). □

This is a finite, classical inertia argument specialized to the exact paired signs in the source. It does not assert that a small residual is zero. The scale on the right is indispensable: for $A = \mathrm{diag}(1, -1)$ and $E = \varepsilon I_2$, the residual norm is ε^2 and $\sigma_{\min}(E)^2 = \varepsilon^2$. Absolute residuals tend to zero, but $d = 2 > p = 1$. The strict threshold correctly fails.

To use enclosures, let $\|\widetilde{E} - E\| \le \eta_E$, $\|\widetilde{A} - A\| \le \eta_A$, and know $\|A\| \le M_A$. Then

$$\|E^T A E\| \le \|\widetilde{E}^T \widetilde{A} \widetilde{E}\| + M_A(2\|\widetilde{E}\|\eta_E + \eta_E^2) + \eta_A\|\widetilde{E}\|^2. \tag{28}$$

Expand around $\widetilde{E}$ while first holding the exact A fixed, then change A to $\widetilde{A}$, to prove the formula. Combine it with (26) and a certified positive lower bound for every $|a_\alpha|$. Rational matrices can alternatively be treated by exact elimination and exact identities. No timing advantage is claimed. Closely spaced real nodes can make the barycentric denominators and numerical condition costs large even though the symbolic detector remains invertible.

The applicability of this proposition to crossing numbers still requires the actual signed support, correct local rotations, polynomial space, and geometric cycle identities. Testing one numerical E, A pair does not prove the source's assertion for every drawing and every n. A coordinate change cannot manufacture these missing objects.

9 Comparison, status, and future work

9.1 What the local study adds

The source already has the exact even-order deletion transfer, the arbitrary-order node detector, the two-page parity count, and the injectivity/isotropy argument. Those are not counted as new contributions here. TR071 already emphasizes parity, integer corrections, and rotation realizability for a different bipartite problem. The distinct local contribution is the exact optimality limit (17) together with the constructive integer extension (18), stated for the complete graph's four-endpoint crossing model. The explicit obstruction shows why that limit is not a drawing theorem. The defect and stability ledgers then identify how much additional information is required.

These results offer an inexpensive rejection test for proposed improvements: if a proposed method uses only the same single-deletion lower bounds and the four-uniform model, it cannot exceed C_*. They also make every parity term and integer ceiling available before a larger computation begins. The limitations are substantial. No new universal crossing constant is obtained; higher-subgraph constraints may still fail to characterize drawings; and a finite numerical margin must not be

mistaken for a topological realization. The near-optimality bounds describe incidence counts, not geometric distance or uniqueness of a minimizer.

Promising future work would impose genuine overlap-consistent rotation data on the four-set model, determine which extra induced constraints exclude the integer extremizers, and independently formalize the supplied normalization and cycle pairing. Exact recovery of a finite certificate is feasible work; convergence of a hierarchy to actual drawings is a separate theorem and is not promised here.

9.2 Primary public status as checked on October 8, 2026

The primary records consulted distinguish several different achievements. The two-page theorem of Ábrego and coauthors proves the exact target within that restricted drawing class [2]. Balogh, Lidický and Salazar obtain an asymptotic lower ratio above 0.985 for unrestricted complete-graph drawings [3]; the supplied source cites a more precise coefficient, but the present literature refresh uses only the primary abstract's conservative statement. Streltsova–Wagner prove the target for spherical shortest-geodesic arc drawings [4], not arbitrary curved plane routes. Aichholzer's institutional publication record reports the computer-assisted exact values at orders 13 and 14 [6]. That computation was not rerun here.

Chikkatur–Zeng's September 10 primary preprint studies Archdeacon's stronger rotation-system conjecture [5]. Its abstract reports verification through order 10, an asymptotic lower bound, and a proof for antipodally shellable systems, not a theorem realizing every rotation system as a simple drawing. This is directly relevant to the distinction between finite rotation constraints and actual geometric realizability. Only the abstract and version metadata were consulted here.

These primary abstracts and records were read for status and scope, not treated as a full external proof audit. They do not independently verify the supplied September 23 claim for every ordinary drawing. Conversely, an older abstract calling the problem open does not disprove a later manuscript. The correct report-level statement is that the source makes a stronger global claim, whose argument is mapped here, while the new local theorems above neither assume publication implies correctness nor claim a separate resolution. The bipartite companion and private comparison do not replace that global obligation.

10 Read scope, exact checks, and assessment

The cached public text was read literally in full: main manuscript 942 lines, bibliography 241 lines, README 20 lines, and three figures of 48, 42, and 48 lines, totaling 1,341 lines. The figure code and captions were read as part of the source architecture; no rendered-source figure audit is claimed. The 486-line private comparison TR071 was also read in full. The original TR248 baseline was freshly read at lines 1–64, 133–175, 349–404, and 1102–1120, a total of 182 distinct lines. Its remaining text and the rest of the private corpus were not read. Heading searches and bibliographic locators are not proof reads.

The owned ledger records pre-read and post-read hashes, exact displayed intervals, and their unread complements. The public files are the pinned cached bytes, separately compared with the parent-provided captured-tree provenance. The Ram file is an existing local baseline copy, not a new transcription produced for this report; no unobserved historical identity with an earlier original is asserted. All confidential material remained local.

The independent author script checks exact parity and induced-subgraph identities; the constructive integer optimum; finite nonnegative weight certificates; the seven-vertex obstruction; explicit two-page alternating pairs; and small rational detector and cycle-pairing fixtures. Cases

and assertions are separately reported in its receipt. These checks do not constitute a proof by exhaustive search: there are 3,846 parameter cases and 110,810 exact assertions, using only integer and rational arithmetic. In particular, the tests do not enumerate all drawings, establish universal geometric normalization, prove realizability of a rotation system, or certify the global Harary–Hill claim. Independent cross-review counts are not included in the author's counts. No foreign checker was executed.

The contribution assessment is C2, independently of applicability L3. The exact optimization and obstruction are useful source-specific diagnostics, but remain modest consequences of classical finite counting, capacity allocation, and linear algebra. No historical priority is claimed for those ingredients. Successful compilation and hash integrity are artifact checks, not mathematical evidence of a new universal theorem.

References

[1] OpenAI, *The crossing number of complete graphs*, September 23, 2026. Supplied pinned manuscript, `build/main.tex` and three figure sources.

[2] B. M. Ábrego, O. Aichholzer, S. Fernández-Merchant, P. Ramos and G. Salazar, *The 2-page crossing number of K_n*, Discrete & Computational Geometry **49** (2013), 747–777. https://arxiv.org/abs/1206.5669.

[3] J. Balogh, B. Lidický and G. Salazar, *Closing in on Hill's conjecture*, SIAM Journal on Discrete Mathematics **33** (2019), 1261–1276. https://arxiv.org/abs/1711.08958.

[4] E. Streltsova and U. Wagner, *Sublevels in arrangements and the spherical arc crossing number of complete graphs*, 2025. https://arxiv.org/abs/2504.07770.

[5] A. Chikkatur and J. Zeng, *Some results on Archdeacon's conjecture for rotation systems*, September 10, 2026. https://arxiv.org/abs/2609.11599.

[6] O. Aichholzer, *Another Small but Long Step for Crossing Numbers:* $\mathrm{cr}(13) = 225$ *and* $\mathrm{cr}(14) = 315$, CCCG 2021, 72–77. Institutional primary record: https://publications.ist.tugraz.at/abstracts/a-aslscn-21/index.html.

[7] M. Subbarao, *Ram Calculus for Turán's Brick-Factory Problem: Rotation Systems, Copositive Certificates, and Symmetry-Reduced Bounds*, private comparison TR-item-071-astra, 2026. Consulted locally in full; its target is $K_{m,n}$, not K_n.

[8] M. Subbarao, *Stable Adaptive Ram Frames and Task-Aware Approximation*, TR248, September 2026. Existing private baseline file `TR248-Stable-Adaptive-Ram-Frames.tex`; selected original proof passages consulted locally.